Quantification and Relational Division in Double-Categorical OLOGs

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Abstract

Working draft of a paper on quantification, modality, data queries and comprehension, all in the context of double categories. A follow-up to [LP25].

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2 Relational Division

Relational division is a query introduced in [Cod72, §2] that behaves essentially as a universal quantifier over given specified values in the column of a table. First we illustrate this operation with a very simple generic example. We may as well work with binary relations $R\colon C \nrightarrow A$ which are of course equivalently monic arrows $\langle d, c\rangle\colon R \rightarrowtail C \times A$. Recall the notation from the reference

$$r[A] = c(r) \text{ where } r \in R$$

and likewise for $r[C] = d(r)$. This is a bit notationally-heavy, but is consistent with the reference and allows for generalizations beyond the following example. Likewise, for any binary relation T, let $g_T(x) = \{y \mid xTy\}$, that is, the set of all y in the target of T related to x under T. Call this the T-orbit of x. Now, for the example, consider the two relational tables R and S with columns (C, A) and (B, D), respectively:

R	
C	**A**
1	a
1	b
1	c
2	a
2	b
3	b

S	
B	**D**
a	2
b	1
b	2
a	3

Thus, for example, $2Ra$ and $2Rb$. Likewise, $aS2$, et cetera. Note that A and B have entries from the same set $\{a, b, c\}$. This is required to perform division. Now, in the notation of the reference, a division query is

$$R[A \div B]S = \{r[C] \mid r \in R \text{ and } S[B] \subset g_R(r[C])\} = \{1, 2\}.$$

We unpack this carefully to illustrate the operation. Now, the possible values of $r[C]$ are the distinct entries of the C-column or R, namely, 1, 2, and 3. Likewise, $S[B] = \{a, b\}$ is the set of all values of the B-column of S. So, there are three sets of the form $g_R(r[C])$, only two of which contain $S[B]$, as in:

$$\begin{aligned}
S[B] &= \{a, b\} \not\subset \{b\} = g_R(3) \\
S[B] &= \{a, b\} \subset \{a, b, c\} = g_R(1) \\
S[B] &= \{a, b\} = \{a, b\} = g_R(2).
\end{aligned}$$

Hence the division query returns $\{1, 2\}$, exactly *the set of all C-values whose orbit contains all the values of the B-column of S*. More prosaically, a C-value is selected by the query iff it is related to everything on the list of B-values. The universal quantification is thus over the set of B-values. The query allows the comparison of separate tables; note that D was not needed to do the query. In the containment relations above, the universally quantified set does not need to completely describe the orbit of any given element satisfying the division query; that is, 1 is related to c, but this is not a B-value. Likewise 3 is related to something on the list, but not everything.

A more concrete example illustrating the same query is the following. Consider the two relational tables below. On the left, rows pair an apothecary with an ingredient sold there; on the right, rows pair an ingredient with an effect produced by the combination of that ingredient with another in a potion.

Apothecary	
Name	**Ingredient**
Arcadia's Cauldron	giant's toe
Arcadia's Cauldron	wheat
Arcadia's Cauldron	void salts
Elgrim's Elixers	giant's toe
Elgrim's Elixers	wheat
The White Phial	wheat

Potion	
Ingredient	**Effect**
giant's toe	fortify health
wheat	fortify health
wheat	lingering damage magicka
giant's toe	damage stamina regen

Note that the second table does not list the other ingredients required to create the potion producing the effect. Our expertise is that any two such ingredients with the same effect will create the desired potion when combined. Now, suppose we want to create a fortify health potion but are to lazy to fight a giant to get his toe. We'd also really like to one-stop-shop (maybe the merchant will give us a deal if we buy in bulk). The orbits are then the inventories:

$$\begin{aligned} g_{\text{Apoth}}(\text{Arcadia's}) &= \{\text{giant's toe, wheat, void salts}\} \\ g_{\text{Apoth}}(\text{Elgrim's}) &= \{\text{giant's toe, wheat}\} \\ g_{\text{Apoth}}(\text{White Phial}) &= \{\text{wheat}\} \end{aligned}$$

The the set of values quantified over is our shopping list: {giant's toe, wheat}. The query is then

$$\text{Apothecary[Ingredient} \div \text{Ingredient]Potion} = \{\text{Name} \mid \{\text{giant's toe, wheat}\} \subset \text{Inventory(Name)}\}$$

which is just to say that we are looking for the proprietors who have all the ingredients on our shopping list. The result is {Arcadia's, Elgrims}. It does not matter that Arcadia's also has void salts; but it does matter that The White Phial only has wheat but no giant's toe.

There is something awkward about these examples inasmuch as we would probably more likely have tables better reflecting our alchemical expertise such as

Apothecary	
Name	**Ingredient**
Arcadia's Cauldron	giant's toe
Arcadia's Cauldron	wheat
Arcadia's Cauldron	void salts
Elgrim's Elixers	giant's toe
Elgrim's Elixers	wheat
The White Phial	wheat

Potion	
Ingredient	**Effect**
giant's toe	fortify health
wheat	fortify health
wheat	lingering damage magicka
giant's toe	damage stamina regen
swamp fungal pod	restore health
void essence	restore health

.

However, division with quantification over the whole the Ingredient column in the Potion table now produces the empty set since no merchant has all those items on hand. And [Cod72, §2.4] recognizes this in the example queries. Namely, example query #8 to find suppliers supplying at least parts 12, 13 and 15 restricts the part # values to only 12, 13 and 15 without ranging over the whole column before executing the division query. This is exactly a *shopping list* kind of example. In other words, technically, the division query is not set up to do selection from a sub-list of a column on its own. A restriction to just the values of interest from the second table is employed implicitly before doing the division query. In the example immediately above, to execute our query, we would filter the rows with the value "fortify health" first and then do the division query with the ingredients column. The flexibility of the division query as written is that it can be applied to any two tables with columns having the same types of values; the expense is that it might not return anything and may require restrictions to produce results in intuitive cases.

3 OLOGs and Instances

What is going on here from a double-categorical OLOG-perspective [LP25] is that we have two basic relations

$$\boxed{\text{Name}} \xrightarrow[\shortmid]{\text{Apothecary}} \boxed{\text{Ingredient}} \qquad\qquad \boxed{\text{Ingredient}} \xrightarrow[\shortmid]{\text{Potion}} \boxed{\text{Effect}}\ .$$

Suppose that these relations are instanced by the two tables at the end of the previous section, thought of as living in $\mathbb{R}\mathsf{el}$. Now, the OLOG generated will be a cartesian equipment. So, given an individual as a global element of a type, say, Arcadia's: $1 \to \boxed{\text{Name}}$, we can form the restriction

$$\begin{tikzcd}
1 \arrow[r, "\text{Arcadia's Inventory}", "\shortmid"{marking}] \arrow[d, "\text{Arcadia's}"'] \arrow[dr, phantom, "\text{res}"] & \boxed{\text{Ingredient}} \arrow[d] \\
\boxed{\text{Name}} \arrow[r, "\text{Apothecary}"', "\shortmid"{marking}] & \boxed{\text{Ingredient}}
\end{tikzcd}$$

which in the instance is precisely the named merchant's inventory. This would play the role of $g_R(x)$ in the previous section. Likewise, our ingredient list for a fortify health potion is schematized by a restriction:

$$\begin{tikzcd}
\boxed{\text{Ingredient}} \arrow[r, "\text{fortify health potion}", "\shortmid"{marking}] \arrow[d] \arrow[dr, phantom, "\text{res}"] & 1 \arrow[d, "\text{fortify health}"] \\
\boxed{\text{Ingredient}} \arrow[r, "\text{Potion}"', "\shortmid"{marking}] & \boxed{\text{Effect}}
\end{tikzcd}$$

These are both examples of "filter" queries discussed in [LP25]. Now, we are trying ultimately to compare the two restrictions, to see if the ingredients for the fortify health potion are in the various inventories. In the examples above the composites

$$1 \xrightarrow[\shortmid]{\text{Arcadia's Inventory}} \boxed{\text{Ingredient}} \xrightarrow[\shortmid]{\text{fortify health potion}} 1$$

and

$$\boxed{\text{Name}} \xrightarrow[\shortmid]{\text{Apothecary}} \boxed{\text{Ingredient}} \xrightarrow[\shortmid]{\text{fortify health potion}} 1$$

are natural to study but neither returns what we want. In the first case, the $\mathbb{R}\mathsf{el}$-composite will return whichever ingredient Arcadia's has on the list; in the second case, the $\mathbb{R}\mathsf{el}$-composite returns whichever merchants have an ingredient on the list. This is a result of the fact that existential quantification over the instance of the middle type is built into composition in $\mathbb{R}\mathsf{el}$.

If we allow ourselves tabulators in the OLOG, much more progress can be made. This allows the creation of types with projections and universal cells, namely,

$$\begin{tikzcd}
& \boxed{\text{Arcadia's Inventory}} \arrow[dl] \arrow[dr] \arrow[d, phantom, "\text{tab}"] & \\
1 \arrow[rr, "\shortmid" marking, "\text{Arcadia's Inventory}"'] & {} & \boxed{\text{Ingredient}}
\end{tikzcd}$$

and then another

$$\begin{tikzcd}
& \boxed{\text{ingredients for fortify health potion}} \arrow[dl] \arrow[dr] \arrow[d, phantom, "\text{tab}"] & \\
\boxed{\text{Ingredient}} \arrow[rr, "\shortmid" marking, "\text{fortify health potion}"'] & {} & 1
\end{tikzcd}.$$

Now, the shopping query is thus to ask whether for each such inventory, the data reflects a putative relationship schematized as

$$\begin{tikzcd}
\boxed{\text{ingredient for fortify health potion}} \arrow[rr, dashed] \arrow[dr] & & \boxed{\text{Arcadia's Inventory}} \arrow[dl] \\
& \boxed{\text{Ingredient}} &
\end{tikzcd}$$

where, in other words, the inventory in question contains the ingredients on the list. Of course this is not quite right, because we'd like to look at all such inventories simultaneously and pick those merchants whose inventories contain all the ingredients. All the inventories will be contained in our instancing of the ingredient type. So, a better way to do this is just to isolate the type of ingredients and view our shopping list as a subtype of the type of ingredients:

$$\begin{tikzcd}
\boxed{\text{ingredient for fortify health potion}} \arrow[d, tail] \\
\boxed{\text{Ingredient}}
\end{tikzcd}$$

This doesn't necessarily need to be monic. But monic arrows are perfectly well axiomatizable in our framework. Perhaps we use the notation to indicate we are thinking of it as a subtype obtained by comprehension [Jac99, §9.3]. Nor does it matter how the subtype comes about at this stage; it could have been obtained as a tabulator as above, or simply required as a primitive in the setup of the OLOG.

What matters is comparing this subtype with the inventories of all the merchants simultaneously. These are implicit in the instancing of the Apothecary relation, so we study the corner

$$\begin{tikzcd}
& \boxed{\text{ingredient for fortify health potion}} \arrow[d, tail] \\
\boxed{\text{Name}} \arrow[r, "\shortmid" marking, "\text{Apothecary}"'] & \boxed{\text{Ingredient}}
\end{tikzcd}$$

It is tempting just to restrict with an identity morphism on the left. This results in a cartesian cell

$$\begin{tikzcd}
\boxed{\text{Name}} \arrow[r, "\text{Merchant with fortify health ingredient}", "|" marking] \arrow[d, equal] \arrow[dr, phantom, "\text{res}"] & \boxed{\text{ingredient for fortify health potion}} \arrow[d, tail] \\
\boxed{\text{Name}} \arrow[r, "\text{Apothecary}"', "|" marking] & \boxed{\text{Ingredient}}
\end{tikzcd}$$

which filters the table for those rows having one of the two ingredients on our list:

Apothecary	
Name	**Ingredient**
Arcadia's Cauldron	giant's toe
Arcadia's Cauldron	wheat
Elgrim's Elixers	giant's toe
Elgrim's Elixers	wheat
The White Phial	wheat

Notice that this gets rid of the unnecessary entry with the value "void salts" but it does not establish our query. Another operation with which our OLOG comes equipped is extension. In $\mathbb{R}\mathsf{el}$ this acts as an image, and formalizes a select query [LP25]. In the present instance we might try an extension

$$\begin{tikzcd}
\boxed{\text{Name}} \arrow[r, "\text{Merchant with fortify health ingredient}", "|" marking] \arrow[d, equal] \arrow[dr, phantom, "\text{ext}"] & \boxed{\text{ingredient for fortify health potion}} \arrow[d, "!"] \\
\boxed{\text{Name}} \arrow[r, "\text{Merchant with fortify health ingredient}"', "|" marking] & 1
\end{tikzcd}$$

which in the data instance is effectively just a list of the three merchants, since each has at least one ingredient. But if we went to the White Phial, we'd be out of luck. This result is owing to the fact that in $\mathbb{R}\mathsf{el}$ the extension is an image computed via *existential quantification.* So, the composite query here of restriction along the subtype, followed by extension has performed an *existential query*, returning each merchant whose inventory contains an ingredient on our list.

Now, this is clearly not the result we want, but it is not a dead-end, for it suggests a path forward. Extensions are essentially left adjoints to restriction. Inasmuch as extension is thought of as existential quantification and restriction is thought of as substitution, we have an analogy of adjunctions

$$\text{ext} \dashv \text{res} \qquad \leftrightarrow \qquad \exists \dashv (-)^*.$$

Now, of course in nice logical situations substitution also has a right adjoint, which is *universal quantification.* This is of course exactly the kind of operation we are seeking to formalize. What we want then is to complete the adjoint analogy

$$\text{ext} \dashv \text{res} \dashv ??? \qquad \leftrightarrow \qquad \exists \dashv (-)^* \dashv \forall$$

in the theater of OLOGs and of double categories generally. Something like this has been studied in the foundational paper [Shu08] under the guise of *coextension.* It was mentioned there, but not studied in-depth due to being not as well behaved as the other two operations. And indeed this is not what we envision in this paper. For coextensions are had by asking for the equipment $\mathbb{D}_1 \to \mathbb{D}_0 \times \mathbb{D}_0$ to be also a *cofibration* which is too strong for what we have in mind. Rather, we take the approach of *fibrational semantics* [Jac99] and ask merely that our equipments $\mathbb{D}_1 \to \mathbb{D}_0 \times \mathbb{D}_0$ have right adjoints to restriction functors and that these satisfy a Beck-Chevalley condition. This is exactly to require *fibered products*. We shall see in more detail how this recovers exactly the queries of interest below in Section 7. For this we first review quantification in more detail.

4 Quantification and Modality

Both universal and existential quantification occurred in some form in the discussion of Section 3. It is well-known that quantification provides left and right adjoints to substitution [MM92]. Working in ordinary set theory, a proposition is a subset $S \subset X$ or perhaps a monic arrow $S \rightarrowtail X$. Such propositions are in 1-1 correspondence with functions $X \to \mathbf{2}$. That $x \in X$ satisfies the proposition $S \subset X$ is written $S(x)$ and means $x \in S$ too. Now, for a set function $f\colon X \to Y$, there is a usual adjoint situation

$$PX \; \underset{\exists_f}{\overset{\forall_f}{\underset{\longrightarrow}{\overset{\longrightarrow}{\xleftarrow{\;f^*\;}}}}} \; PY \qquad\qquad \exists_f \dashv f^* \dashv \forall_f$$

where

$$\exists_f(S) = \{y \in Y \mid f(x) = y \text{ for some } x \in S\}$$
$$\forall_f(S) = \{y \in Y \mid \text{ for all } x \in X \text{ if } f(x) = y \text{ then } x \in S\}.$$

Of course ordinary quantification is a special case illustrated by taking $f = \pi\colon X \times Y \to Y$, the usual projection to the second factor of the cartesian product. In this case, the quantified subsets are $R \subset X \times Y$, that is, binary relations. That x and y are R-related could be written xRy or perhaps $x \leq_R y$ or $x \sim_R y$. Quantification is then:

$$\exists_\pi(R) = \{y \in Y \mid R(x, y) \text{ for some } x \in X\}$$

$$\forall_\pi(R) = \{y \in Y \mid R(x, y) \text{ for all } x \in X\}.$$

Typically, the left variable is bound. Note also in particular that if $Y = 1$, we have closed formulas *viz.* truth values.

Now, a relation $R\colon X \nrightarrow Y$ might also be thought of as set of pairs of entities and observations or attribute values, or when $X = Y$ as a non-deterministic system. Fixing x and quantifying over y instead suggests, due to non-symmetry, that y is somehow subsequent to, or a consequence of x. This comports with the system view of R that y is a later state accessible from x. Perhaps x is prior to y in a temporal sequence. On the causal view, somehow y is a consequence of x. Now, the set of all $x \in X$ for which some subsequent state or attribute value satisfies a proposition $S \subset Y$ is written

$$\Diamond S = \{x \mid S(y) \text{ for some } x \leq_R y\}.$$

This is exactly the modal possibility operator familiar from the topological semantics given by the Alexandrov Topology. Likewise there is the set of those elements of X, all of whose subsequent states or attribute values satisfy $S \subset Y$, written

$$\Box S = \{x \mid S(y) \text{ for all } x \leq_R y\}.$$

This is modal necessity in the same topological semantics. Thus, in the former case, an element of $\Diamond S$ is an element of X for which S is possible, in the sense that it is true of some later accessible state. In the latter case, an element of $\Box S$ is an element of X for which S is necessarily true, in the sense that any subsequent accessible state satisfies S.

So far this is just the usual $\mathbb{R}\mathsf{el}$-semantics of quantification and modality. These operators recover those discussed in Section 3 and several others. If we go back to the table instancing the relation

Apothecary	
Name	**Ingredient**
Arcadia's Cauldron	giant's toe
Arcadia's Cauldron	wheat
Arcadia's Cauldron	void salts
Elgrim's Elixers	giant's toe
Elgrim's Elixers	wheat
The White Phial	wheat

$$\boxed{\text{Name}} \xrightarrow{\text{Apothecary}} \boxed{\text{Ingredient}}$$

and our list S again has wheat and giant's toe, then directly from the operations above we have six queries summarized in Figure 1. Use R to denote the relation and $R^{\dagger}$ is the reverse relation. The variables x and y range over names and over ingredients, respectively. Notice some curiosities. The first is that R is supposed to be given. Thus, the existential queries are just the select column queries returning either all the merchants or all the ingredients specified by R. These are already studied in [LP25]. The universal queries are new and more interesting, but neither returns our list query since S does not even enter into the construction. The modal queries as well are new. The possibility query returns the same as the existential query only as an accident of R. If The White Phial was paired with something other than wheat, for example, the results would be different. Note as well that the necessity query is not the list query we are looking for. It returns the set of those merchants having only the ingredients on our list. This is a query with utility, but as far as our shopping list goes it does not matter whether the returned merchant has other items. In fact, suppose that one closer has our ingredients but also a lot of others. We would naturally prefer to go there.

Result of Existential Query

$$\begin{aligned}\exists_\pi(R) &= \{y \mid R(x,y) \text{ for some } x \in X\}\\ &= \{\text{ingredients someone has}\}\\ &= \{\text{wheat, giant's toe, void salts}\}\end{aligned}$$

Result of Universal Query

$$\begin{aligned}\forall_\pi(R) &= \{y \mid R(x,y) \text{ for all } x \in X\}\\ &= \{\text{ingredients everyone has}\}\\ &= \{\text{wheat}\}\end{aligned}$$

Result of Reverse Existential Query

$$\begin{aligned}\exists_\pi(R^\dagger) &= \{x \mid R(x,y) \text{ for some } y \in Y\}\\ &= \{\text{those with an } R\text{-ingredient}\}\\ &= \{\text{wheat, giant's toe, void salts}\}\end{aligned}$$

Result of Reverse Universal Query

$$\begin{aligned}\forall_\pi(R^\dagger) &= \{x \mid R(x,y) \text{ for all } y \in Y\}\\ &= \{\text{those with all } R\text{-ingredients}\}\\ &= \{\text{Arcadia's}\}\end{aligned}$$

Result of Possibility Query

$$\begin{aligned}\Diamond S &= \{x \mid S(y) \text{ for some } x \leq_R y\}\\ &= \{x \mid x \text{ has an ingredient on } S\}\\ &= \{\text{Arcadia's, Elgrim's, Phial}\}\end{aligned}$$

Results of Necessity Query

$$\begin{aligned}\Box S &= \{x \mid S(y) \text{ for all } x \leq_R y\}\\ &= \{x \mid x \text{ has only ingredients on } S\}\\ &= \{\text{Elgrim's}\}\end{aligned}$$

Figure 1: Results of six quantification queries.

It will turn out that the list query is given by a pair of operations, one of which is universal quantification, applied in sequence. And so, to enrich the datascheme to handle this, the goal now is to formulate all of the above operations purely double-categorically and axiomatize them in our generated OLOGs. Ultimately, we will allow the data-instancing and double-categorical functorial semantics to perform the desired queries.

5 Existential Queries

The set-up we are entertaining is that of having a subtype $s\colon S \rightarrowtail Y$ thought of as specifying in the associated data some subcollection or list of items in the ambient list instancing Y. We have already seen that in $\mathbb{R}\mathsf{el}$, merely composing has the effect of an existential query. This can be formalized in any equipment $\mathbb{D}$ by first viewing S as an extension and then composition with R on the left as in

$$\begin{tikzcd} & S \arrow[r, "\mathrm{id}_S", "\shortmid"{marking}] \arrow[d, tail, "s"'] \arrow[dr, phantom, "\mathrm{ext}"] & S \arrow[d, "!"] \\ X \arrow[r, "\shortmid"{marking}, "R"'] & Y \arrow[r, "\shortmid"{marking}, "S"'] & 1 \end{tikzcd}.$$

In $\mathbb{R}\mathsf{el}$, the result $R \otimes S$ is precisely $\Diamond S$ as above. An equivalent $\mathbb{R}\mathsf{el}$-construction is to form the pullback on the left followed by the image

$$\begin{tikzcd} R \odot s^* \arrow[r, tail] \arrow[d, tail] \arrow[dr, phantom, "\lrcorner", very near start] & R \arrow[d, tail] \\ X \times S \arrow[r, tail, "1 \times s"'] & X \times Y \end{tikzcd} \quad \rightsquigarrow \quad \begin{tikzcd} R \odot s^* \arrow[r, tail] \arrow[dr, two heads] & X \times S \arrow[r, "1 \times !"] & X \times 1 \\ & \Diamond S \arrow[ur, tail] & \end{tikzcd}$$

That is, the corner of the pullback on the right is exactly the set of pairs (x, a) where $x \sim_R a$ under R. The image then binds $x \in X$ via existential quantification. This is formalized in any equipment $\mathbb{D}$ by first taking a restriction and then an extension:

$$\begin{tikzcd} X \arrow[r, "R \odot u^*", "\shortmid"{marking}] \arrow[d, "1"'] \arrow[dr, phantom, "\mathrm{res}"] & S \arrow[d, tail, "s"] \\ X \arrow[r, "\shortmid"{marking}, "R"'] & X \end{tikzcd} \quad \rightsquigarrow \quad \begin{tikzcd} X \arrow[r, "R \odot u^*", "\shortmid"{marking}] \arrow[d, "1"'] \arrow[dr, phantom, "\mathrm{ext}"] & S \arrow[d, "!"] \\ X \arrow[r, "\shortmid"{marking}, "\Diamond S"'] & 1 \end{tikzcd}.$$

Now, in any equipment $\mathbb{D}$ there is a comparison cell of the form

$$\begin{tikzcd} X \arrow[r, "R \odot s^*", "\shortmid"{marking}] \arrow[d, "1"'] \arrow[dr, phantom, "\mathrm{res}"] & S \arrow[r, "\mathrm{id}", "\shortmid"{marking}] \arrow[d, "s"] \arrow[dr, phantom, "\mathrm{ext}"] & S \arrow[d, "!"] \\ X \arrow[r, "\shortmid"{marking}, "R"'] & X \arrow[r, "\shortmid"{marking}, "S"'] & 1 \end{tikzcd}$$

Thus, there is a globular cell $\Diamond S \Rightarrow R \odot S$ since $\Diamond S$ is constructed as an extension:

$$\begin{tikzcd} X \arrow[rr, "R \odot s^*", "\shortmid"{marking}] \arrow[d] \arrow[drr, phantom, "\cong"] & & S \arrow[d] \\ X \arrow[r, "R \odot s^*", "\shortmid"{marking}] \arrow[d, "1"'] \arrow[dr, phantom, "\mathrm{res}"] & S \arrow[r, "\mathrm{id}", "\shortmid"{marking}] \arrow[d, "s"] \arrow[dr, phantom, "\mathrm{ext}"] & S \arrow[d, "!"] \\ X \arrow[r, "\shortmid"{marking}, "R"'] & X \arrow[r, "\shortmid"{marking}, "S"'] & 1 \end{tikzcd} \quad = \quad \begin{tikzcd} X \arrow[r, "R \odot u^*", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\mathrm{ext}"] & S \arrow[d, "!"] \\ X \arrow[r, "\shortmid"{marking}, "\Diamond S"'] \arrow[d, equal] \arrow[dr, phantom, "\exists !"] & 1 \arrow[d, equal] \\ X \arrow[r, "\shortmid"{marking}, "R \odot S"'] & 1 \end{tikzcd} \tag{5.1}$$

Now, it can be seen directly that this is an invertible cell when $\mathbb{D}$ is either $\mathbb{R}\mathsf{el}$ or $\mathbb{S}\mathsf{pan}$ or $\mathbb{M}\mathsf{at}(\mathsf{V})$ for suitably structured monoidal V. The case of $\mathbb{R}\mathsf{el}$ is trivial, so we treat the other two.

Example 5.1. In $\mathbb{S}\mathsf{pan}$, work with $\langle d, c\rangle \colon R \to X \times Y$ and then on the one hand

$$\Diamond S = \{((x, a), r) \mid x \in X, a \in S, r \in R, d(r) = x, c(r) = a\}$$

whereas $R \odot \hat{s}$ is the corner of the pullback:

$$R \times_X S = \{(r, a) \mid r \in R, a \in S, c(r) = a\}.$$

These are evidently isomorphic, as the latter just forgets the indices $x \in X$. But these are recoverable as $d(r) = x$. □

Example 5.2. Let V denote any monoidal category with coproducts preserved by the tensor. We then have the double category $\mathbb{M}\mathsf{at}(\mathsf{V})$ of V-matrices. Restrictions and extensions are computed as

$$\begin{tikzcd} Z \arrow[r, "f_! M g^*", "\shortmid"{marking}] \arrow[d, "f"'] \arrow[dr, phantom, "\text{res}"] & W \arrow[d, "g"] \\ X \arrow[r, "M"', "\shortmid"{marking}] & Y \end{tikzcd} \qquad f_! M g^*(z,w) = M(fz, gw)$$

and

$$\begin{tikzcd} Z \arrow[r, "N", "\shortmid"{marking}] \arrow[d, "f"'] \arrow[dr, phantom, "\text{ext}"] & W \arrow[d, "g"] \\ X \arrow[r, "f^* N g_!"', "\shortmid"{marking}] & Y \end{tikzcd} \qquad f^* N g_!(x,y) = \sum_{\substack{fz=x \\ gw=y}} A(z,w)$$

Now, to form the possibility operator, we have first the restriction

$$\begin{tikzcd} X \arrow[r, "M \odot s^*", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\text{res}"] & S \arrow[d, "s"] \\ X \arrow[r, "M"', "\shortmid"{marking}] & X \end{tikzcd} \qquad (M \odot s^*)(x,a) = M(x,a) \qquad \text{for } x \in X, a \in S$$

and then via an extension we have the possibility operator:

$$\begin{tikzcd} X \arrow[r, "M \odot s^*", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\text{ext}"] & S \arrow[d, "!"] \\ X \arrow[r, "\Diamond S"', "\shortmid"{marking}] & ! \end{tikzcd} \qquad \Diamond S(x) = \sum_{a \in S} M(x,a) \qquad \text{for } x \in X.$$

Now, to compare this to the composite $M \odot S$ note first that S viewed as a proarrow $S \colon X \nrightarrow 1$ has the entries described by

$$\begin{tikzcd} S \arrow[r, "\mathrm{id}_S", "\shortmid"{marking}] \arrow[d, "s"'] \arrow[dr, phantom, "\text{ext}"] & S \arrow[d, "!"] \\ X \arrow[r, "S"', "\shortmid"{marking}] & 1 \end{tikzcd} \qquad S(x) = \begin{cases} I & x \in S \\ 0 & \text{o/w} \end{cases} \qquad \text{for } x \in X.$$

This follows from the definition of the unit proarrows in $\mathbb{M}\mathsf{at}(\mathsf{V})$. Now, from the definition of the loose composite of two matrices, we have

$$\begin{aligned} (M \odot S)(x) &= \sum_{y \in X} M(x,y) \otimes S(y) \\ &= \sum_{y \in X} M(x,y) \otimes \begin{cases} I & y \in U \\ 0 & \text{o/w} \end{cases} \\ &= \sum_{y \in S} M(x,y) \end{aligned}$$

for any $x \in X$, showing that the two constructions are the same in this case too. $\square$

Now, in fact the two construction agree in general for any equipment such as $\mathbb{P}\mathsf{rof}$ or $\mathbb{M}\mathsf{at}(\mathsf{V})$ for more complicated V, each of which is difficult to treat directly.

Proposition 5.3. *The globular cell $\Diamond S \Rightarrow R \odot S$ induced in Equation* (5.1) *is an isomorphism in any equipment. Accordingly, the two possible definitions of $\Diamond S$ agree.*

Proof. It suffices to show that the two sides of Equation (5.1) compute the same extension. But this is just a result of how they are computed in equipments. That is, S viewed as a proarrow defined via the extension is the composite $S := s^* \odot \mathrm{id}_S \odot !_S$ where $!_S$ is the companion of $!\colon S \to 1$. So, the composite is $R \odot S = R \odot s^* \odot \mathrm{id}_S \odot !_S$. Likewise, $\Diamond S$ as the extension of $R \odot s^*$ along 1_X and $!$ is just $\Diamond S = R \odot s^* \odot !_S$. These are clearly isomorphic by inserting appropriate associators and units coming from the equipment structure. More abstractly, the left side of Equation (5.1) can thus be rearranged to make $R \odot S$ exhibit the same extension explicitly. □

So, although the constructions are thus essentially the same, we make a choice as to which is basic. We insist on starting with the subtype $s\colon S \to Y$ consistent with the set-up of Section 3. And so in either case there are two operations to perform the query. There is a slight semantic advantage to the latter. For in the case of $\mathbb{S}\mathsf{pan}$, although the appropriate element $x \in X$ is implicit in the composite as $d(r) = x$, our preference is to keep the data as a triple $((x, a), r) = (x, a, r)$ as in the second way of computing the possibility operator. This way all the information is recorded in the apex of the span and the two legs can be forgotten.

Definition 5.4 (Possibility)**.** Let $s\colon S \rightarrowtail X$ denote any monic arrow in the equipment $\mathbb{D}$. The proposition **possibly** S is then formed as an restriction followed by an extension:

$$\begin{tikzcd} X \arrow[r, "R\odot u^*", "\shortmid"{marking}] \arrow[d, "1"'] \arrow[dr, phantom, "\text{res}"] & S \arrow[d, tail, "s"] \\ X \arrow[r, "R"', "\shortmid"{marking}] & X \end{tikzcd} \quad \rightsquigarrow \quad \begin{tikzcd} X \arrow[r, "R\odot u^*", "\shortmid"{marking}] \arrow[d, "1"'] \arrow[dr, phantom, "\text{ext}"] & S \arrow[d, "!"] \\ X \arrow[r, "\Diamond S"', "\shortmid"{marking}] & 1 \end{tikzcd}$$

as in the pattern suggested above: □

Note then that the content of Proposition 5.3 is thus that of a *rewrite rule* for the possibility operator and thus also for existential queries of this kind.

6 Fibrational Semantics I

All the instances of existential quantification introduced above are instances of *fibrational semantics* of the equipment $\mathbb{D}$. We have the general picture:

$$\Sigma_{f,g}\colon \mathbb{D}(A, B) \rightleftarrows \mathbb{D}(X, Y)\colon (f, g)^* \qquad \Sigma_{f,g} \dashv (f, g)^* \tag{6.1}$$

for any arrows $f\colon A \to X$ and $g\colon B \to Y$ of $\mathbb{D}$. The hom categories are the fibers of $\langle \mathrm{src}, \mathrm{tgt}\rangle\colon \mathbb{D}_1 \to \mathbb{D}_0 \times \mathbb{D}_0$ over the pairs of objects (A, B) and (X, Y). Here $\Sigma_{f,g}$ denotes the left adjoint to substitution/restriction along f and g. This is provided automatically since $\langle \mathrm{src}, \mathrm{tgt}\rangle$ is both a fibration and opfibration [Shu08]. Of course $\Sigma_{f,g}$ is thus extension in the equipment structure.

Both instances of existential quantification are recoverable in this framework. On the one hand, for $s\colon S \to Y$ we have the functors

$$\mathbb{D}(X, Y) \xrightarrow{(1_X, s)^*} \mathbb{D}(X, S) \xrightarrow{\Sigma_S} \mathbb{D}(X, 1) \tag{6.2}$$

and consequently

$$\Sigma_S(1_X, s)^*(R) = \Diamond S$$

as in Definition 5.4. On the other hand, ordinary existential quantification is just an application of a left adjoint:

$$\mathbb{D}(X,Y) \xrightarrow{\Sigma_X} \mathbb{D}(1,Y) \qquad \Sigma_X(R) = \exists x.R(x,y)$$

as has already been observed in the case of $\mathbb{R}\mathsf{el}$. In each case, the left adjoints used are special cases of *Kan quantification* [Law02]. Let $f\colon A \to X$ denote any morphism of $\mathbb{D}$. Consider the *one-sided* adjunction

$$\Sigma_f\colon \mathbb{D}(A,Y) \rightleftarrows \mathbb{D}(X,Y)\colon f^* \qquad \Sigma_f \dashv f^* \tag{6.3}$$

Owning to the nature of the computation of restrictions and extensions, the two functors are simply: *precompose with a companion or conjoint* as in $\Sigma_f = f^* \odot -$ and $f^* = f_! \odot -$ and the adjunction is then

$$f^* \odot -\colon \mathbb{D}(A,Y) \rightleftarrows \mathbb{D}(X,Y)\colon f_! \odot - \qquad f^* \odot - \dashv f_! \odot - \tag{6.4}$$

Retaining the Σ-notation for the left adjoint, the adjunction thus expresses an existential quantification introduction rule. Namely, it presents a bijection between globular cells

$$\begin{tikzcd} X \arrow[r, "\Sigma_f R", "|"] \arrow[d, equal] \arrow[dr, phantom, "\Downarrow"] & Y \arrow[d, equal] \\ X \arrow[r, "S"', "|"] & Y \end{tikzcd} \quad \leftrightarrow \quad \begin{tikzcd} A \arrow[rr, "R", "|"] \arrow[d, equal] \arrow[drr, phantom, "\Downarrow"] & & Y \arrow[d, equal] \\ A \arrow[r, "f_!"', "|"] & X \arrow[r, "S"', "|"] & Y \end{tikzcd} \qquad \frac{\Sigma_f R \Rightarrow S}{R \Rightarrow f_! \odot S}.$$

With the computation $\Sigma_f = f^* \odot -$ as above, we have an analogue of the computation of the left adjoint to substitution for V-functors in [Law02, §3]. Applying the bijection to the identity on $\Sigma_f R$, there is a single cell

$$\begin{tikzcd} A \arrow[rr, "R", "|"] \arrow[d, equal] \arrow[drr, phantom, "\Downarrow"] & & Y \arrow[d, equal] \\ A \arrow[r, "f_!"', "|"] & X \arrow[r, "\Sigma_f R"', "|"] & Y \end{tikzcd}$$

and the universal property makes $\Sigma_f R$ into a (left) *Kan extension* of R along $f_!$ in the underlying bicategory of $\mathbb{D}$. Specializing to the case of Σ_X as above and writing the traditional '$\exists$' for the left adjoint, the bijection expressing the universal property is precisely the usual existential introduction rule:

$$\begin{tikzcd} X \arrow[r, "\exists_X R", "|"] \arrow[d, "!"'] \arrow[dr, phantom, "\Downarrow"] & Y \arrow[d, equal] \\ 1 \arrow[r, "S"', "|"] & Y \end{tikzcd} \quad \leftrightarrow \quad \begin{tikzcd} X \arrow[rr, "R", "|"] \arrow[d, equal] \arrow[drr, phantom, "\Downarrow"] & & Y \arrow[d, equal] \\ X \arrow[r, "X_!"', "|"] & 1 \arrow[r, "S"', "|"] & Y \end{tikzcd} \qquad \frac{\exists_X R \Rightarrow S}{R \Rightarrow X_! \odot S}$$

Notice that this construction provides a left Kan extension in the underlying bicategory of $\mathbb{D}$ and thus that $\mathbb{D}$ is in this sense bicategorically closed. This is not a fragment of the closed structure studied in [Shu08].

7 Universal Queries

Now we turn to universal quantification starting from fibrational semantics and Kan quantification. That the restriction functor $(f,g)^*\colon \mathbb{D}(x,y) \to \mathbb{D}(a,b)$ has a right adjoint is the statement that there exists a functor

$$\prod_{f,g}\colon \mathbb{D}(a,b) \to \mathbb{D}(x,y)$$

and natural isomorphisms

$$\mathbb{D}(a,b)(f_! m g^*, s) \cong \mathbb{D}(x,y)(m, \prod_{f,g}(s))$$

establishing natural families of bijections of globular cells

$$\frac{f_! m g^* \Rightarrow s}{m \Rightarrow \prod_{f,g}(s)}$$

given in each direction by application of the appropriate functor and then composition with a unit or counit as the case may be. Notice again that this specializes to a bidirectional universal quantification rule. The elementary characterization of this right adjoint as a universal in this context is the following.

Proposition 7.1. *The restriction functor $(f,g)^*\colon \mathbb{D}(x,y) \to \mathbb{D}(a,b)$ has a right adjoint if, and only if, for each $s\colon a \nrightarrow b$ there is a specified proarrow $\forall_{f,g}(s)$ together with a globular cell ϵ_s of the form*

$$\begin{tikzcd} a \arrow[r, "f_!(\forall_{f,g}(s))g^*", "\shortmid"{marking}] \arrow[d, "f"'] \arrow[dr, phantom, "res"] & b \arrow[d, "g"] \\ x \arrow[r, "\forall_{f,g}(s)"', "\shortmid"{marking}] & y \end{tikzcd} \quad \rightsquigarrow \quad \begin{tikzcd} a \arrow[r, "f_!(\forall_{f,g}(s))g^*", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\epsilon_s"] & b \arrow[d, equal] \\ a \arrow[r, "s"', "\shortmid"{marking}] & b \end{tikzcd}$$

such that for any other globular cell from a restriction along f and g, say, of the form

$$\begin{tikzcd} a \arrow[r, "f_! w g^*", "\shortmid"{marking}] \arrow[d, "f"'] \arrow[dr, phantom, "res"] & b \arrow[d, "g"] \\ x \arrow[r, "w"', "\shortmid"{marking}] & y \end{tikzcd} \quad \rightsquigarrow \quad \begin{tikzcd} a \arrow[r, "f_! w g^*", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\xi"] & b \arrow[d, equal] \\ a \arrow[r, "s"', "\shortmid"{marking}] & b \end{tikzcd}$$

there is a unique cell $\hat{\xi}$ for which h factors through ϵ_s as in

$$\begin{tikzcd} a \arrow[r, "f_! w g^*", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\hat{\xi}"] & b \arrow[d, equal] \\ a \arrow[r, "f_!(\forall_{f,g}(s))g^*", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\epsilon_s"] & b \arrow[d, equal] \\ a \arrow[r, "s"', "\shortmid"{marking}] & b \end{tikzcd} \quad = \quad \begin{tikzcd} a \arrow[r, "f_! w g^*", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\xi"] & b \arrow[d, equal] \\ a \arrow[r, "s"', "\shortmid"{marking}] & b \end{tikzcd}$$

Proof. This is just the characterization of an adjunction in terms of the existence of a counit as a universal ([Mac98, §IV.1, Theorem 2(iv)]). Note that naturality of ϵ_s in s follows automatically. □

Remark 7.2. There is a crucial conceptual and technical difference here in the fibrational semantics *vis-á-vis* the prior case of existential quantification. This is that universal quantification is not neatly expressible via loose composition with a companion or conjoint; nor is there some obvious third type of proarrow f_* associated to a morphism f with its own accompanying base-change cells and universal properties that would do this. For although equipments $\mathbb{D}$ involve both a fibration and opfibration $\langle \mathrm{src}, \mathrm{tgt}\rangle \colon \mathbb{D}_1 \to \mathbb{D}_0 \times \mathbb{D}_0$, this same functor is not usually also a cofibration [Shu08]. This would the structure required to provide some kind of *corestriction* as a right adjoint to restriction. Universal quantification thus must appear under some different guise. Our approach of course has been that of fibrational semantics: adjoints to substitution are the basic notion; that in one case they happen to be computed by companions or conjoints is a nice feature of equipments, but evidently does not carry over the others. Now, our axiomatized notion is the following, based on [Jac99, §1.9]. This is a geometric/fibrational analogue of a *quantification hyperdoctrine* [Law06], [Law70] instantiated here for double categories. As defined in [LP25], a double OLOG is a small and finitely-presented 'double category of relations' [Lam22]. This is in particular a locally posetal cartesian equipment. □

Definition 7.3. A **quantification double OLOG** is a double OLOG where each substitution functor has a right adjoint satisfying Beck-Chevalley. □

Example 7.4. Certainly $\mathbb{R}\mathsf{el}$ has such adjoints since they are inherited from Set and computed in the manner described in the introduction. Beck-Chevalley follows or can be verified directly. Thus, given an appropriate additional *reverse* operator on proarrows, we have an axiomatization of the two universal queries of Figure 1 given by application of the appropriate right adjoint to whatever proarrow is under consideration. □

Our main application of universal quantification is, however, the list-query described Section 2 where we look for those proprietors having all the items on our shopping list. This can be done by a composition of a restriction followed by a universal quantifier right adjoint. Start with a given proarrow $R \colon X \nrightarrow Y$ and a subtype $s \colon S \to Y$ representing our list. Consider then the composite functor

$$\mathbb{D}(X, Y) \xrightarrow{(1_X, s)^*} \mathbb{D}(X, S) \xrightarrow{\prod_{1_X, !}} \mathbb{D}(X, 1) \tag{7.1}$$

assuming of course that such right adjoints exist. Note that this is a direct analogue of the composite Equation (6.2) which produced the modal query retrieving exactly those proprietors with some item on the list S. In the case of relations, this has the following effect.

Example 7.5. Specializing Equation (7.1) to $\mathbb{R}\mathsf{el}$, recall that the restriction $R \odot s_!$ is the set of those pairs (x, y) such that $x \sim_R y$ and $y \in S$. Now, the map $X \times S \to X \times 1$ is up to bijection between codomains the same as the first projection $X \times S \to X$. So, using the computation of $\forall_{\pi_X}$ in Section 4, we have

$$\begin{aligned}
\forall_{\pi_X}(R \odot s_!) &= \{x \in X \mid \text{ for all } (x', y) \in X \times S\, (\pi(x', y) = x \text{ implies } (x', y) \in R \odot s_!)\} \\
&= \{x \in X \mid \text{ for all } (x, y) \in X \times S\, (x, y) \in R \odot s_!\} \\
&= \{x \in X \mid \text{ for all } y \in S\, (x, y) \in R \odot s_!\} \\
&= \{x \in X \mid \text{ for all } y \in S\, (x \sim_R y)\}
\end{aligned}$$

That is, the result of the application of the composite of functors to R is the set of those X-elements x for which everything in S is R-related to x. This is precisely to say that if S is a list of ingredients and X is our list of proprietors, then the query returns the list of those who have everything on the list (but could have other items). □

It is worth pointing out that although this construction thus achieves the desired effect and in this sense the objective of the paper too, it leaves a gap in the narrative inasmuch as it is not a modal query like the existential query relative to S that it mimics. The explanation is the relative flexibility of the existential operator. That is, existential validity asks just that *something* satisfy the quantified formula, relation or proposition. This is a relatively weak requirement. And so the result of the possibility query in Figure 1 is just the existential query $\exists$ applied to $R \odot s_!$. Universal queries, by their nature surveying all entities under the scope of the quantifier, are more stringent. Thus, $\forall(R \odot s_!)$ and $\Box S$ produce dramatically different results: the first case that of the shops with all ingredients on the list; and second those with only ingredients on the list. That is, the first case quantifies over list items and checks against inventory; the second quantifies over inventory and checks against list items. The difficulty consists in the fact that what needs to be returned in each case is the list of proprietors. The built-in for $\forall$ as a right adjoint must match variances without the mixing this would demand in one or the other case. That is, the comprehension of $\forall$ must always match the codomain typing of operator as a right adjoint. The object $R \odot s_!$ must be formed to compare shops with out list items. But just, for example, switching the projection or reversing the relation results in the codomain typing of $\forall$ to S-objects which will return ingredients, not shops in the query. To recover this operator we turn to tabulators.

8 Tabulators and Modality

Definition 8.1 ([GP99])**.** A double category $\mathbb{D}$ **has tabulators** if external identity has a right adjoint

$$\mathbb{D}_0 \underset{T}{\overset{\mathrm{id}_-}{\rightleftarrows}} \mathbb{D}_1 \qquad\qquad \mathrm{id}_- \dashv T.$$

The **tabulator** of a proarrow $m\colon x \nrightarrow y$ is a cell τ_m having the universal property that there exists a unique morphism h making an equality of diagrams

$$\begin{tikzcd} Tm \arrow[r, equal, "\shortmid" marking] \arrow[d, "d"'] \arrow[dr, phantom, "\tau_m"] & Tm \arrow[d, "c"] \\ x \arrow[r, "m"', "\shortmid" marking] & y \end{tikzcd} \quad = \quad \begin{tikzcd} Tm \arrow[r, equal, "\shortmid" marking] \arrow[d, "h"'] \arrow[dr, phantom, "\mathrm{id}_h"] & Tm \arrow[d, "h"] \\ z \arrow[r, equal, "\shortmid" marking] \arrow[d, "f"'] \arrow[dr, phantom, "\theta"] & z \arrow[d, "g"] \\ x \arrow[r, "m"', "\shortmid" marking] & y \end{tikzcd}$$

for each such cell θ. $\Box$

9 Structure on Relations

This operator justly bears the '$\Diamond$' notation not only on account of the semantics discussed above. There is also the following observation, namely, that what is the case is possibly the case, as in the usual monadic interpretation of possibility. The proof is most naturally interpreted in $\mathbb{R}\mathsf{el}$, but is completely formal. When $\mathbb{R}\mathsf{el}$ is meant, the relationship is the familiar $S \leq \Diamond S$.

Proposition 9.1. *If R is reflexive, then there is a globular cell $S \Rightarrow \Diamond S$.*

Proof. Reflexivity of R means that there is a canonical cell

$$\begin{tikzcd}
X \arrow[r, "\mathrm{id}_X", "|"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\delta_X"] & X \arrow[d, equal] \\
X \arrow[r, "R"', "|"{marking}] & X
\end{tikzcd}$$

expressing the fact that the diagonal factors through R. Consequently, since $R \odot u^*$ arises as a restriction, there is a unique cell making an equality

$$\begin{tikzcd}
S \arrow[r, "\mathrm{id}_S", "|"{marking}] \arrow[d, "s"'] \arrow[dr, phantom, "\mathrm{id}_s"] & S \arrow[d, "s"] \\
X \arrow[r, "\mathrm{id}_X", "|"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\delta_X"] & X \arrow[d, equal] \\
X \arrow[r, "R"', "|"{marking}] & X
\end{tikzcd}
=
\begin{tikzcd}
S \arrow[r, "\mathrm{id}_S", "|"{marking}] \arrow[d, "s"'] \arrow[dr, phantom, "\exists\,!"] & S \arrow[d, equal] \\
X \arrow[r, "R\odot s^*", "|"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\mathrm{res}"] & S \arrow[d, "s"] \\
X \arrow[r, "R"', "|"{marking}] & X
\end{tikzcd}$$

And so, putting this cell as (I) in the diagram below, we get the desired comparison induced by the universal property of the extension cell:

$$\begin{tikzcd}
S \arrow[r, "\mathrm{id}_S", "|"{marking}] \arrow[d, "s"'] \arrow[dr, phantom, "(\mathrm{I})"] & S \arrow[d, equal] \\
X \arrow[r, "R\odot s^*", "|"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\mathrm{ext}"] & S \arrow[d, "!"] \\
X \arrow[r, "\Diamond S"', "|"{marking}] & 1
\end{tikzcd}
=
\begin{tikzcd}
S \arrow[r, "\mathrm{id}_S", "|"{marking}] \arrow[d, "s"'] \arrow[dr, phantom, "\mathrm{ext}"] & S \arrow[d, "!"] \\
X \arrow[r, "S", "|"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\exists\,!"] & 1 \arrow[d, equal] \\
X \arrow[r, "\Diamond S"', "|"{marking}] & 1
\end{tikzcd}$$

Notice the abuse of notation, namely, that we think of S as both the object providing the domain of s but also as a proarrow $S\colon X \nrightarrow 1$ obtained by extension. ☐

Part II
Cartesian Closedness & Full First-Order Logic

10 Modularity and Frobenius

It is asserted in [Lam22, §3] that any 'double category of relations' satisfies the usual *Frobenius Law* for the accompanying doctrine. This is proved here in detail since Frobenius the proof is interesting in its own right and the law is the main ingredient in proving that right adjoints to substitution do give cartesian closed structure, locally.

Recall from [CW87] that any 'bicategory of relations' is compact closed and has an involution law

$$f \mapsto f^\circ$$

satisfying

$$1^\circ = 1 \qquad (f^\circ)^\circ = f \qquad (gf)^\circ = f^\circ g^\circ$$

with $(-)^\circ$ defined using the unit and counit in the compact closed structure as in the proof of [CW87, Theorem 2.4]. Now, if f is a *map* then f° is that right adjoint $f \dashv f^\circ$ [CW87, Lemma 2.5]. This extends to 'double categories of relations' in the following way.

Lemma 10.1. *In any 'double category of relations', the equations*

1. $(f_!)^\circ = f^*$
2. $(f^*)^\circ = f_!$

each hold.

Proof. Start with the usual equipment adjunction $f_! \dashv f^*$ [Shu08]. The underlying bicategory of any 'double category of relations' is a 'bicategory of relations' [Lam22] and bicategorical adjoints are unique up to isomorphism. $\square$

The proof of the Modular Law (cf. [FS90]) is now an application. It is phrased here in the manner needed in proving Frobenius.

Proposition 10.2 (Modular Laws)**.** *In any 'double category of relations', the two inequalities*

1. $(f^* \odot m) \wedge n \leq f^* \odot (m \wedge f_! \odot n)$
2. $(p \odot f_!) \wedge q \leq (p \wedge q \odot f^*) \odot f_!$

each hold.

Proof. Two inequalities follow directly from [CW87, Theorem 2.4]. These are

1. $\Delta_! \odot (f^* \times 1) \leq f^* \odot \Delta_! \odot (1 \times f_!)$
2. $(f_! \times 1) \odot \Delta^* \leq (1 \times f^*) \odot \Delta^* \odot f_!$

using $(f^*)^\circ = f_!$ as in the lemma above for the first. The two modular laws are direct applications. For the first one, we have

$$\begin{aligned}(f^* \odot m) \wedge n &= \Delta_! \odot (f^* \odot m \times n) \odot \Delta^* \\ &= \Delta_! \odot (f^* \times 1 \odot m \times n) \odot \Delta^* \\ &\leq f^* \odot \Delta_! \odot (1 \times f_! \odot m \times n) \odot \Delta^* \\ &= f^* \odot \Delta_! \odot (m \times f_! \odot n) \odot \Delta^* \\ &= f^* \odot (m \wedge f_! \odot n)\end{aligned}$$

using the fact the interaction of $\odot$ and $\times$ as well as the definition of local products. The other inequality is analogous. $\square$

Construction 10.3 (Frobenius Comparison Cell)**.** Let $\mathbb{D}$ denote any cartesian equipment. Local products make sense in this context. Let $f\colon x \to z$ and $g\colon y \to w$ denote morphisms. First form the coproduct extension as the third step in the sequence:

$$\begin{tikzcd} x \arrow[r, "f_!pg^*", "\shortmid"{marking}] \arrow[d, "f"'] \arrow[dr, phantom, "\mathrm{res}"] & y \arrow[d, "g"] \\ z \arrow[r, "p"', "\shortmid"{marking}] & w \end{tikzcd} \quad \rightsquigarrow \quad \begin{tikzcd} x \arrow[r, "m\wedge f_!pg^*", "\shortmid"{marking}] \arrow[d, "\Delta_x"'] \arrow[dr, phantom, "\mathrm{res}"] & y \arrow[d, "\Delta_y"] \\ x \times x \arrow[r, "m\times f_!pg^*"', "\shortmid"{marking}] & y \times y \end{tikzcd} \quad \rightsquigarrow \quad \begin{tikzcd} x \arrow[r, "m\wedge f_!pg^*", "\shortmid"{marking}] \arrow[d, "f"'] \arrow[dr, phantom, "\mathrm{ext}"] & y \arrow[d, "g"] \\ z \arrow[r, "\coprod_{f,g}(m\wedge f_!pg^*)"', "\shortmid"{marking}] & w \end{tikzcd} .$$

Note that we are suppressing some of the loose composition notation for readability. Now, on the other hand, form the local product as the second step in the sequence:

$$\begin{tikzcd} x \arrow[r, "m"] \arrow[d, "f"'] \arrow[dr, phantom, "\text{ext}"] & y \arrow[d, "g"] \\ z \arrow[r, "\coprod_{f,g} m"'] & w \end{tikzcd} \quad \rightsquigarrow \quad \begin{tikzcd} z \arrow[r, "(\coprod_{f,g} m)\wedge p"] \arrow[d, "\Delta_z"'] \arrow[dr, phantom, "\text{res}"] & w \arrow[d, "\Delta_w"] \\ z \times z \arrow[r, "(\coprod_{f,g} m)\times p"'] & w \times w \end{tikzcd}$$

By the universal property of the local product immediately above, there is thus a canonical comparison cell

$$m \wedge f_!pg^* \Rightarrow (\coprod_{f,g} m) \wedge p$$

and likewise a comparison cell

$$\coprod_{f,g}(m \wedge f_!pg^*) \Rightarrow (\coprod_{f,g} m) \wedge p$$

by the universal property of the extension cell defining the coproduct in the preantepenultimate display. This is the required *Frobenius comparison cell.* Note that it exists even if, as assumed, $\mathbb{D}$ is merely a cartesian equipment. □

Proposition 10.4. *Any 'double category of relations' satisfies the Frobenius Law. That is, the canonical cell*

$$\coprod_{f,g}(m \wedge f_!pg^*) \leq (\coprod_{f,g} m) \wedge p$$

described above is invertible.

Proof. For the proof, we produce an reverse inequality. Uniqueness then forces the inequality above to be an equality. Starting from the left side and using the construction of extensions and both Modular Laws (Proposition 10.2), we have

$$\begin{aligned} (\coprod_{f,g} m) \wedge p &= (f^* \odot m \odot g_!) \wedge p \\ &\leq f^* \odot (m \odot g^* \wedge f_! \odot p) \\ &\leq f^* \odot (m \wedge (f_! \odot p \odot g^*)) \odot g_! \\ &= \coprod_{f,g}(m \wedge f_!pg^*) \end{aligned}$$

as required. □

Query Example

11 Cartesian Closed Structure

If $\mathbb{D}$ is a cartesian equipment, then each hom-category $\mathbb{D}(x, y)$ of course has cartesian products, given by the formula

$$\begin{tikzcd} x \arrow[r, "1_{x,y}"] \arrow[d, "!"'] \arrow[dr, phantom, "\text{res}"] & y \arrow[d, "!"] \\ 1 \arrow[r] & 1 \end{tikzcd} \qquad \begin{tikzcd} x \arrow[r, "m\wedge n"] \arrow[d, "\delta_x"'] \arrow[dr, phantom, "\text{res}"] & y \arrow[d, "\delta_y"] \\ x \times x \arrow[r, "m\times n"'] & y \times y \end{tikzcd}$$

as described in [Ale18, Proposition 4.3.2].

Definition 11.1. A cartesian equipment $\mathbb{D}$ is **locally cartesian closed** if each product functor

$$m \wedge - : \mathbb{D}(X, Y) \to \mathbb{D}(X, Y)$$

has a right adjoint $(-)^m$. The object n^m is called the **exponential** of n by m. □

Example 11.2. For relations $R, S : X \nrightarrow Y$, the exponential is given by

$$S^R = \{(x, y) \mid xRy \text{ implies } xSy\}$$

and computed via adjoints as

$$\forall_R R^*(S)$$

thinking of R as its own tabulator. □

Of course we may simply ask that our cartesian equipments are locally cartesian closed in this sense. It is a point of interest, however, to connect this structure to the right adjoints to substitution assumed in previous sections inasmuch as, (1) these are supposed to provide *generalized hom objects* and (2) there is an evident universal quantification over pairs (x, y) in the example above. The answer, roughly, is that these right adjoints are computable from right adjoints to substitution whenever the given cartesian equipment has strong tabulators

Construction 11.3. Let $\mathbb{D}$ denote any cartesian equipment with tabulators. Let $\mathbb{S}\mathsf{pan}(\mathbb{D}_0)(x, y)$ denote the ordinary category of spans in $\mathbb{D}_0$ from x to y. Note that we do not yet assume the existence of suitable pullbacks making $\mathbb{S}\mathsf{pan}(\mathbb{D}_0)$ into a double category. The assignment

$$\int_{x,y} (-) : \mathbb{D}(x, y) \to \mathbb{S}\mathsf{pan}(\mathbb{D}_0)(x, y) \qquad \text{by} \qquad m \quad \mapsto \quad (x \leftarrow Tm \to y)$$

is functorial by the universal property of tabulators. Likewise, the assignment taking a pairing morphism to its extension

$$\mathrm{fib}_{x,y} : \mathbb{S}\mathsf{pan}(\mathbb{D}_0)(x, y) \to \mathbb{D}(x, y) \qquad \text{by} \qquad x \xleftarrow{l} z \xrightarrow{r} y \quad \mapsto \quad \begin{tikzcd} z \arrow[r, equals, "|" marking] \arrow[d, "l"'] \arrow[dr, phantom, "\text{ext}"] & z \arrow[d, "r"] \\ x \arrow[r, "l^* \odot r_!"', "|" marking] & y \end{tikzcd}$$

is a functor by the universal property of extensions. Note that these are essentially the hom-localizations of the double functors giving the oplax/lax span representations of [Nie12]. For our purposes, it is straightforward to check that each such $\mathrm{fib}_{x,y}$ is a left adjoint to the corresponding integral functor

$$\mathrm{fib}_{x,y} \dashv \int_{x,y} (-)$$

again by universal properties of the involved constructions. Note that the integral functor, being a right adjoint, preserves limits. As observed in [Lam22, Lemma 6.3] the dashed comparison

$$\begin{tikzcd} x \arrow[r, "\langle 1,1 \rangle"] \arrow[d, dashed, "\eta"'] & x \times x \arrow[d, equals] \\ T\,\mathrm{id}_x \arrow[r] & x \times x \end{tikzcd}$$

is an isomorphism iff $\mathrm{id}_- : \mathbb{D}_0 \to \mathbb{D}_1$ is fully faithful, that is, if $\mathbb{D}$ is unit-pure [Ale18, §4.3.7]. Now, this is □

Remark 11.4. Tabulators in this sense provide a *comprehension scheme*, that is, one half of an elements-style correspondence whose pseudo-inverse is a *fibers construction*, associating, for example in the case of discrete (op)fibrations, the (co)presheaf whose values at a given object is the precisely the fiber of the (op)fibration over that object. Note that we have imposed no conditions on the receiving category of the elements-style tabulator functors other than that it be a slice category. This is so as not to prejudice the *type* of fibration concept the elements construction recovers. See [Lam22, §8] for more discussion on these points. These functors organize into lax transformations in the following way. □

Proposition 11.5. *If $\mathbb{D}$ is a cartesian equipment with tabulators, the fibers and integral functors developed above are components of respective lax natural transformations*

$$\int(-)\colon \mathbb{D}(-,=) \Rightarrow \mathbb{S}\mathsf{pan}(\mathbb{D}_0)(-,=) \qquad \text{and} \qquad \mathrm{fib}\colon \mathbb{S}\mathsf{pan}(\mathbb{D}_0)(-,=) \Rightarrow \mathbb{D}(-,=)$$

of indexed categories (equivalently opfibrations) forming a fibered adjunction.

Proof. x □

The next definition say that elements and fibers correctly roundtrip in the direction that makes sense, namely, first tabulating and then extending results in the original proarrow. The other roundtrip does not make sense to axiomatize since we do not specify the fibration type.

Definition 11.6. The tabulator of a proarrow $p\colon X \nrightarrow Y$ is **strong** if the canonical morphism

$$\begin{tikzcd} Tp \arrow[r, equal, "\shortmid" marking] \arrow[d, "d"'] \arrow[dr, phantom, "\tau_p"] & Tp \arrow[d, "c"] \\ x \arrow[r, "p"', "\shortmid" marking] & y \end{tikzcd} \quad = \quad \begin{tikzcd} Tp \arrow[r, equal, "\shortmid" marking] \arrow[d, "d"'] \arrow[dr, phantom, "\tau_p"] & Tp \arrow[d, "c"] \\ x \arrow[r, "d^*\odot c_!"', "\shortmid" marking] \arrow[d, equal] \arrow[dr, phantom, "\exists!"] & y \arrow[d, equal] \\ x \arrow[r, "p"', "\shortmid" marking] & y \end{tikzcd}$$

is an isomorphism. A double category has **strong tabulators** if it has tabulators and each is strong. □

Remark 11.7. This means, equivalently, that each such tabulator cell τ_p is an extension cell. Consequently, a double category has strong tabulators if each of its proarrows is the extension of its tabulator. The double category of relations is an example. □

Definition 11.8. Tabulators are **cartesian** if the counit transformation $\tau\colon \mathrm{id}_{-}T \Rightarrow 1$ is a cartesian transformation. □

As notation, for $m\colon x \nrightarrow y$ and its tabulator Tm, we write $m^*(n)$ for the proarrow domain of the restriction cell

$$\begin{tikzcd} Tm \arrow[r, "m^*(n)", "\shortmid" marking] \arrow[d, "d"'] \arrow[dr, phantom, "\mathrm{res}"] & Tm \arrow[d, "c"] \\ x \arrow[r, "n"', "\shortmid" marking] & y \end{tikzcd}$$

Likewise $\prod_m$ and Σ_m denote the right- and left-adjoints to substitution along the domain and codomain morphisms coming with Tm.

Theorem 11.9. *If $\mathbb{D}$ is a unit-pure 'double category of relations' with $\prod$, then each hom category $\mathbb{D}(x, y)$ is cartesian closed and*

$$m \wedge n \cong \coprod_m (\mathrm{id}_{Tm} \wedge m^*(n)) \qquad\qquad n^m \cong \prod_m m^*(n)$$

compute the product and exponentials.

Proof. □

Remark 11.10. The proof is inspired by the argument of [Jac99, Proposition 10.5.4]. We shall show that local products are given by extensions. The second statement will follow since right adjoints are unique up to isomorphism. Consider:

$$\begin{aligned}
\mathbb{D}_1(x, y)(p, \coprod_m m^*(n)) &\cong \mathbb{D}_0/x \times y(t_p, t_{\coprod_m m^*(n)}) && \text{(strength)}\\
&\cong \mathbb{D}_0/x \times y(t_p, \Sigma_m t_{m^*(n)}) && \text{(cocontinuity)}\\
&\cong \mathbb{D}_0/x \times y(t_p, t_m \times t_n) && \text{(cartesianness)}\\
&\cong \mathbb{D}_0/x \times y(t_p, t_m) \times \mathbb{D}(p, t_n)\\
&\cong \mathbb{D}_1(p, m) \times \mathbb{D}_1(p, n)
\end{aligned}$$

which shows that $\Sigma_m m^*(n)$ computes the cartesian product $m \wedge n$ in $\mathbb{D}(x, y)$. Since $\prod_m m^*$ is the right adjoint, this proves that $\mathbb{D}(x, y)$ is cartesian closed. □

Corollary 11.11. *If a cartesian equipment $\mathbb{D}$ with $\prod$-quantification satisfying Beck-Chevalley has tabulators that are*

1. *strong*
2. *cartesian*
3. *cocontinuous*

then $\mathbb{D}$ is locally cartesian closed as a fibration.

Proof. Reindexing preserves the closed structure by BC. □

12 Cocartesianness and Negation

13 Modal FOL

14 Prospectus: The Role of Compactness

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